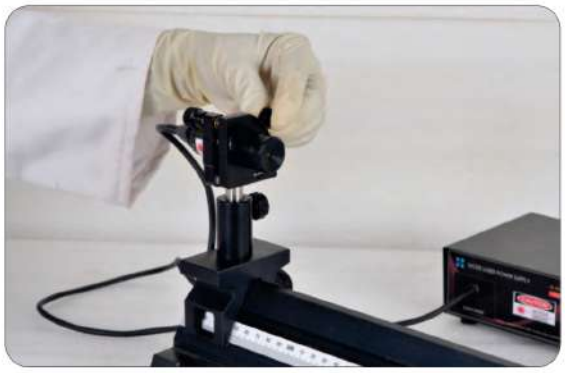
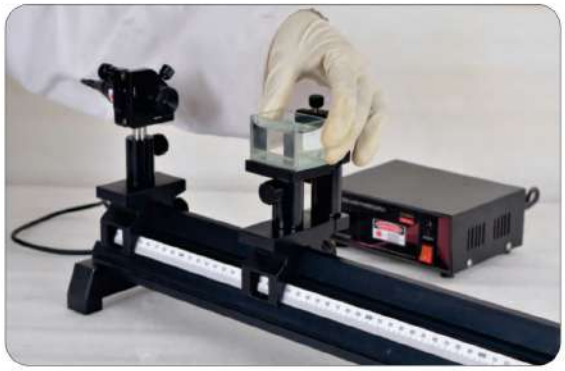
Experimental Set-up

1. Fix the laser mount on optical rail.

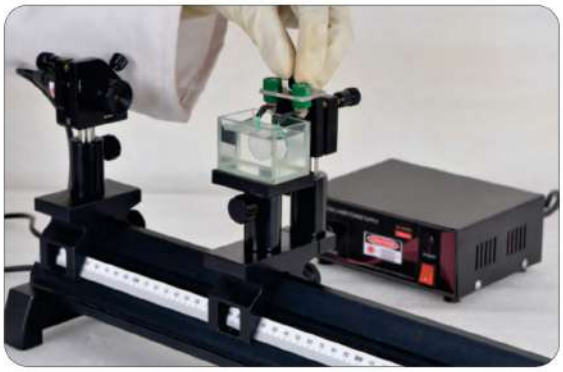
2. Place laser on the mount properly.

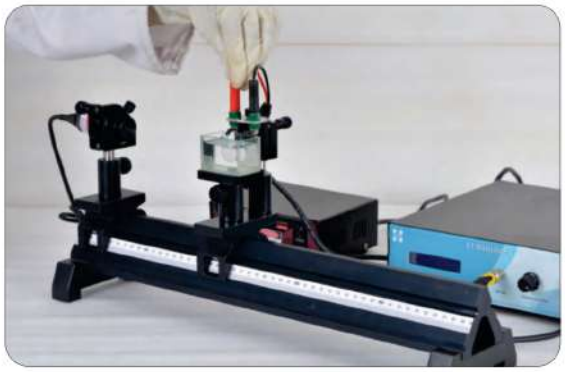


3. Place the glass tank holder on rail. Fill the glass tank with liquid and keep it on the tank holder.



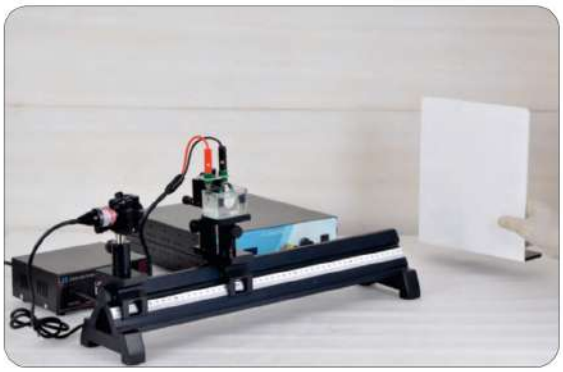
4. Fix the crystal on the mount and keep it immersed fully in the liquid. .
Connect it to the RF oscillator.

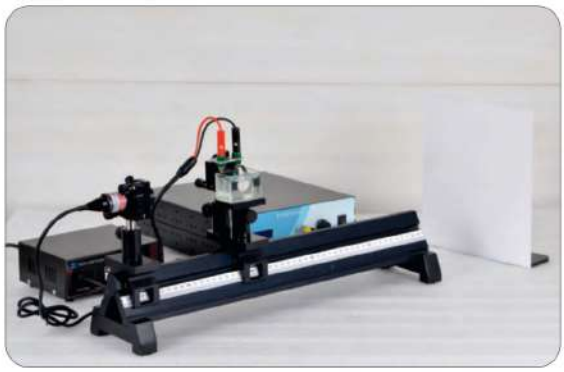




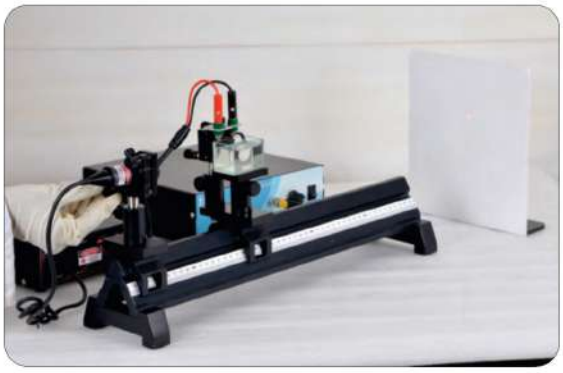
5. Place the screen in front of the glass tank at a particular distance.

NB: The distance from the glass tank to the screen should be at least 2.5meters for obtaining a clear pattern.



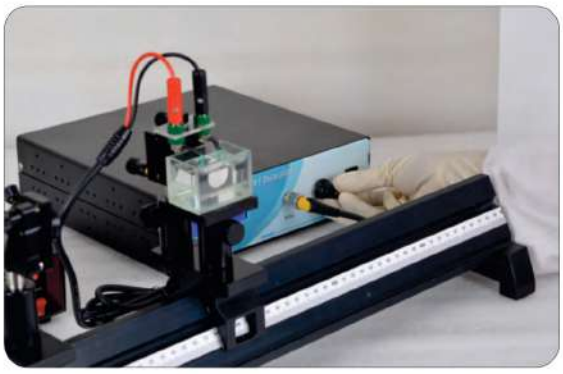


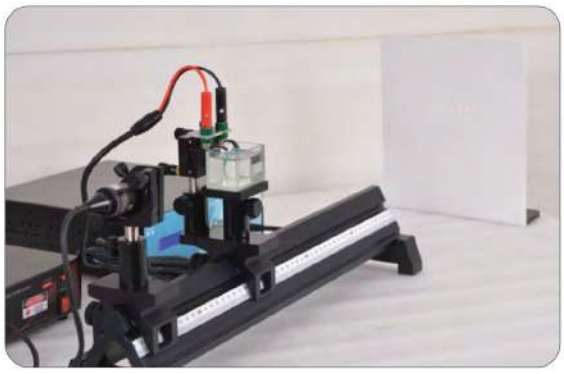
6. Switch on the laser and RF oscillator. Align the crystal and laser so that laser beam is parallel to the face of the crystal. Adjust the kinematic setup provided on the laser mount to get the beam in the field of standing wave generated.

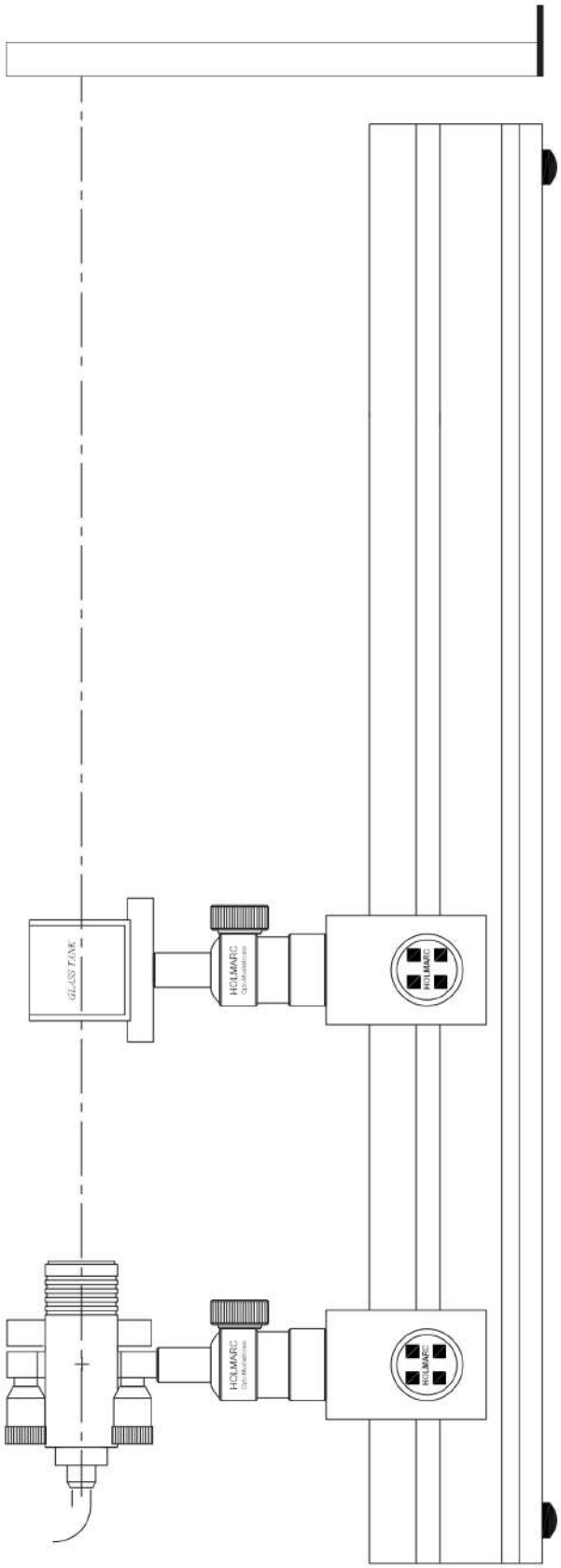


Note: *Make sure that there are no air bubbles in the liquid especially around the crystal.*

7. Keep the laser spot falling on screen and adjust the frequency of the oscillator until you get a very good fringe pattern on both sides of the central bright spot.





8. Using a graph paper we can plot and measure the distance from the central bright spot to the nth order spot.

Mechanical Drawing